Analysis on BGL

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2025-02-13

Data Loading

The data are stored in Book1.xlsx file in both wide format and long format. [1] The R program will load the long format in Sheet 2. The structure of the data is with 3 columns: Treatments, Duration and Blood Glucose Level (BGL) [2]

```
library(readxl)
library(tidyverse)
Book1 <- read_excel("Book1.xlsx",
                    sheet = "Sheet2")
```

Data Preparation

using [3]. First of all, ([2]) the treatments names along with the control are extracted and stored in variable "trts". Then the BGL values at 45 minutes are taken as the initial reference points for comparison with other BGL of later times. BGL at 0 minutes are not included in the analysis cause this is the assessment data before the administration of both treatments and BGL induction. ([1])

```
trts <- Book1 %>% distinct(Treatments) %>% pull(Treatments)
trts
```

```
[1] "Negative Control" "Positive Control" "Normal"           "SakarPhyu(0.5 g)"
[5] "SakarPhyu(1 g)"   "SakarPhyu(1.5 g)"
```

```
initial <- Book1 %>% filter(time==45) %>% pull(BGL)
names(initial)<-rep(trts, each = 5)

fasting<-Book1 %>% filter (time == 0)
# print(fasting, n = Inf)
df<-Book1 %>% filter(time!=0) %>% mutate(Percent_reduction=case_when(
  Treatments==trts[1] ~ (BGL-initial[1:5])/initial[1:5]*100,
  Treatments==trts[2] ~ (BGL-initial[6:10])/initial[6:10]*100,
  Treatments==trts[3] ~ (BGL-initial[11:15])/initial[11:15]*100,
  Treatments==trts[4] ~ (BGL-initial[16:20])/initial[16:20]*100,
  Treatments==trts[5] ~ (BGL-initial[21:25])/initial[21:25]*100,
  Treatments==trts[6] ~ (BGL-initial[26:30])/initial[26:30]*100
))
```

	Treatments	time	BGL	Percent_reduction
1	Negative Control	45	225	0.0000000
2	Negative Control	45	165	0.0000000
3	Negative Control	45	153	0.0000000
4	Negative Control	45	180	0.0000000
5	Negative Control	45	201	0.0000000
6	Negative Control	90	183	-18.6666667
7	Negative Control	90	160	-3.0303030
8	Negative Control	90	137	-10.4575163
9	Negative Control	90	170	-5.5555556
10	Negative Control	90	185	-7.9601990
11	Negative Control	135	158	-29.7777778
12	Negative Control	135	153	-7.2727273
13	Negative Control	135	127	-16.9934641
14	Negative Control	135	167	-7.2222222
15	Negative Control	135	168	-16.4179104
16	Negative Control	180	134	-40.4444444
17	Negative Control	180	129	-21.8181818
18	Negative Control	180	108	-29.4117647
19	Negative Control	180	118	-34.4444444
20	Negative Control	180	157	-21.8905473
21	Negative Control	225	147	-34.6666667
22	Negative Control	225	115	-30.3030303
23	Negative Control	225	109	-28.7581699
24	Negative Control	225	133	-26.1111111
25	Negative Control	225	149	-25.8706468
26	Positive Control	45	168	0.0000000
27	Positive Control	45	190	0.0000000
28	Positive Control	45	209	0.0000000
29	Positive Control	45	189	0.0000000
30	Positive Control	45	152	0.0000000
31	Positive Control	90	93	-44.6428571
32	Positive Control	90	108	-43.1578947
33	Positive Control	90	153	-26.7942584
34	Positive Control	90	123	-34.9206349
35	Positive Control	90	107	-29.6052632
36	Positive Control	135	65	-61.3095238
37	Positive Control	135	85	-55.2631579
38	Positive Control	135	93	-55.5023923
39	Positive Control	135	123	-34.9206349
40	Positive Control	135	92	-39.4736842
41	Positive Control	180	89	-47.0238095
42	Positive Control	180	114	-40.0000000
43	Positive Control	180	80	-61.7224880
44	Positive Control	180	110	-41.7989418
45	Positive Control	180	107	-29.6052632
46	Positive Control	225	84	-50.0000000
47	Positive Control	225	91	-52.1052632
48	Positive Control	225	80	-61.7224880
49	Positive Control	225	107	-43.3862434
50	Positive Control	225	126	-17.1052632
51	Normal	45	120	0.0000000
52	Normal	45	123	0.0000000

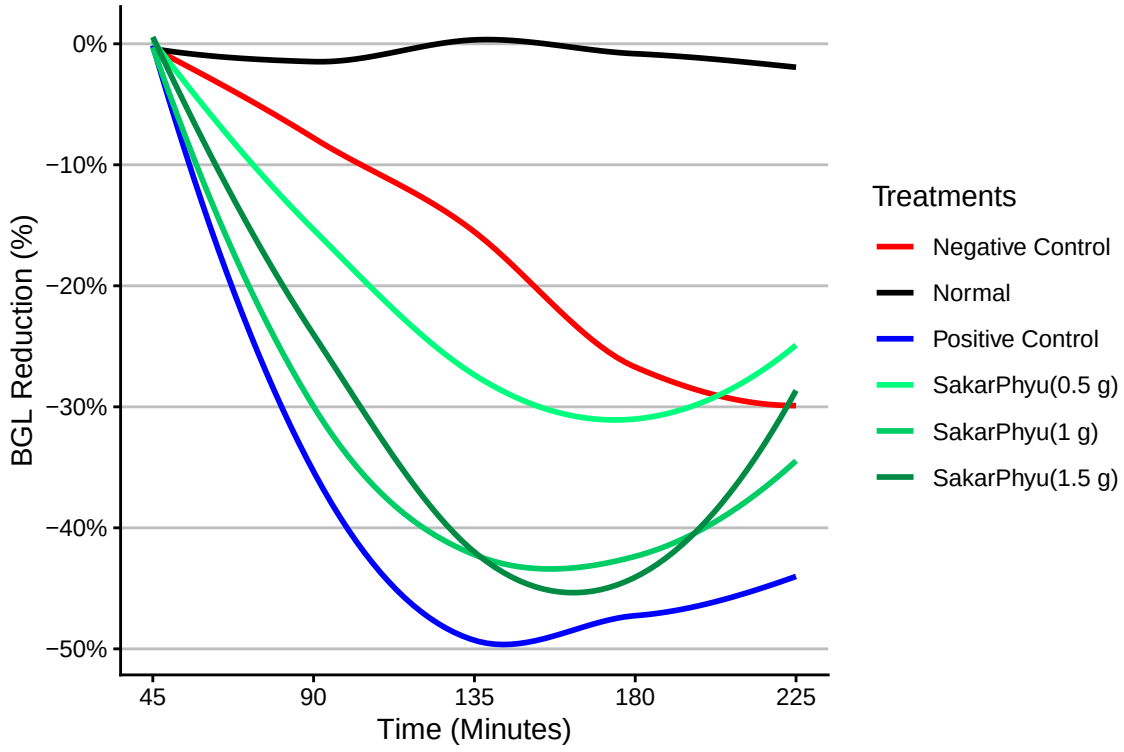
	Treatments	time	BGL	Percent_reduction
53	Normal	45	118	0.0000000
54	Normal	45	119	0.0000000
55	Normal	45	118	0.0000000
56	Normal	90	117	-2.5000000
57	Normal	90	120	-2.4390244
58	Normal	90	115	-2.5423729
59	Normal	90	118	-0.8403361
60	Normal	90	110	-6.7796610
61	Normal	135	119	-0.8333333
62	Normal	135	125	1.6260163
63	Normal	135	120	1.6949153
64	Normal	135	121	1.6806723
65	Normal	135	115	-2.5423729
66	Normal	180	118	-1.6666667
67	Normal	180	120	-2.4390244
68	Normal	180	117	-0.8474576
69	Normal	180	120	0.8403361
70	Normal	180	112	-5.0847458
71	Normal	225	121	0.8333333
72	Normal	225	119	-3.2520325
73	Normal	225	115	-2.5423729
74	Normal	225	118	-0.8403361
75	Normal	225	115	-2.5423729
76	SakarPhyu(0.5 g)	45	180	0.0000000
77	SakarPhyu(0.5 g)	45	141	0.0000000
78	SakarPhyu(0.5 g)	45	189	0.0000000
79	SakarPhyu(0.5 g)	45	166	0.0000000
80	SakarPhyu(0.5 g)	45	190	0.0000000
81	SakarPhyu(0.5 g)	90	147	-18.3333333
82	SakarPhyu(0.5 g)	90	130	-7.8014184
83	SakarPhyu(0.5 g)	90	163	-13.7566138
84	SakarPhyu(0.5 g)	90	110	-33.7349398
85	SakarPhyu(0.5 g)	90	194	2.1052632
86	SakarPhyu(0.5 g)	135	104	-42.2222222
87	SakarPhyu(0.5 g)	135	145	2.8368794
88	SakarPhyu(0.5 g)	135	142	-24.8677249
89	SakarPhyu(0.5 g)	135	63	-62.0481928
90	SakarPhyu(0.5 g)	135	170	-10.5263158
91	SakarPhyu(0.5 g)	180	89	-50.5555556
92	SakarPhyu(0.5 g)	180	140	-0.7092199
93	SakarPhyu(0.5 g)	180	139	-26.4550265
94	SakarPhyu(0.5 g)	180	72	-56.6265060
95	SakarPhyu(0.5 g)	180	150	-21.0526316
96	SakarPhyu(0.5 g)	225	105	-41.6666667
97	SakarPhyu(0.5 g)	225	141	0.0000000
98	SakarPhyu(0.5 g)	225	150	-20.6349206
99	SakarPhyu(0.5 g)	225	90	-45.7831325
100	SakarPhyu(0.5 g)	225	159	-16.3157895
101	SakarPhyu(1 g)	45	221	0.0000000
102	SakarPhyu(1 g)	45	157	0.0000000
103	SakarPhyu(1 g)	45	206	0.0000000
104	SakarPhyu(1 g)	45	238	0.0000000

	Treatments	time	BGL	Percent_reduction
105	SakarPhyu(1 g)	45	172	0.0000000
106	SakarPhyu(1 g)	90	118	-46.6063348
107	SakarPhyu(1 g)	90	117	-25.4777070
108	SakarPhyu(1 g)	90	105	-49.0291262
109	SakarPhyu(1 g)	90	168	-29.4117647
110	SakarPhyu(1 g)	90	164	-4.6511628
111	SakarPhyu(1 g)	135	100	-54.7511312
112	SakarPhyu(1 g)	135	133	-15.2866242
113	SakarPhyu(1 g)	135	80	-61.1650485
114	SakarPhyu(1 g)	135	117	-50.8403361
115	SakarPhyu(1 g)	135	122	-29.0697674
116	SakarPhyu(1 g)	180	121	-45.2488688
117	SakarPhyu(1 g)	180	137	-12.7388535
118	SakarPhyu(1 g)	180	80	-61.1650485
119	SakarPhyu(1 g)	180	99	-58.4033613
120	SakarPhyu(1 g)	180	119	-30.8139535
121	SakarPhyu(1 g)	225	135	-38.9140271
122	SakarPhyu(1 g)	225	141	-10.1910828
123	SakarPhyu(1 g)	225	95	-53.8834951
124	SakarPhyu(1 g)	225	93	-60.9243697
125	SakarPhyu(1 g)	225	156	-9.3023256
126	SakarPhyu(1.5 g)	45	122	0.0000000
127	SakarPhyu(1.5 g)	45	182	0.0000000
128	SakarPhyu(1.5 g)	45	149	0.0000000
129	SakarPhyu(1.5 g)	45	192	0.0000000
130	SakarPhyu(1.5 g)	45	166	0.0000000
131	SakarPhyu(1.5 g)	90	118	-3.2786885
132	SakarPhyu(1.5 g)	90	113	-37.9120879
133	SakarPhyu(1.5 g)	90	87	-41.6107383
134	SakarPhyu(1.5 g)	90	172	-10.4166667
135	SakarPhyu(1.5 g)	90	139	-16.2650602
136	SakarPhyu(1.5 g)	135	74	-39.3442623
137	SakarPhyu(1.5 g)	135	132	-27.4725275
138	SakarPhyu(1.5 g)	135	67	-55.0335570
139	SakarPhyu(1.5 g)	135	69	-64.0625000
140	SakarPhyu(1.5 g)	135	126	-24.0963855
141	SakarPhyu(1.5 g)	180	69	-43.4426230
142	SakarPhyu(1.5 g)	180	113	-37.9120879
143	SakarPhyu(1.5 g)	180	57	-61.7449664
144	SakarPhyu(1.5 g)	180	91	-52.6041667
145	SakarPhyu(1.5 g)	180	126	-24.0963855
146	SakarPhyu(1.5 g)	225	63	-48.3606557
147	SakarPhyu(1.5 g)	225	134	-26.3736264
148	SakarPhyu(1.5 g)	225	64	-57.0469799
149	SakarPhyu(1.5 g)	225	194	1.0416667
150	SakarPhyu(1.5 g)	225	145	-12.6506024

Visualization

Use package developed by [4]. The **percent change** will be plotted against "time" and the data for each treatment will be presented as separate colored smooth lines

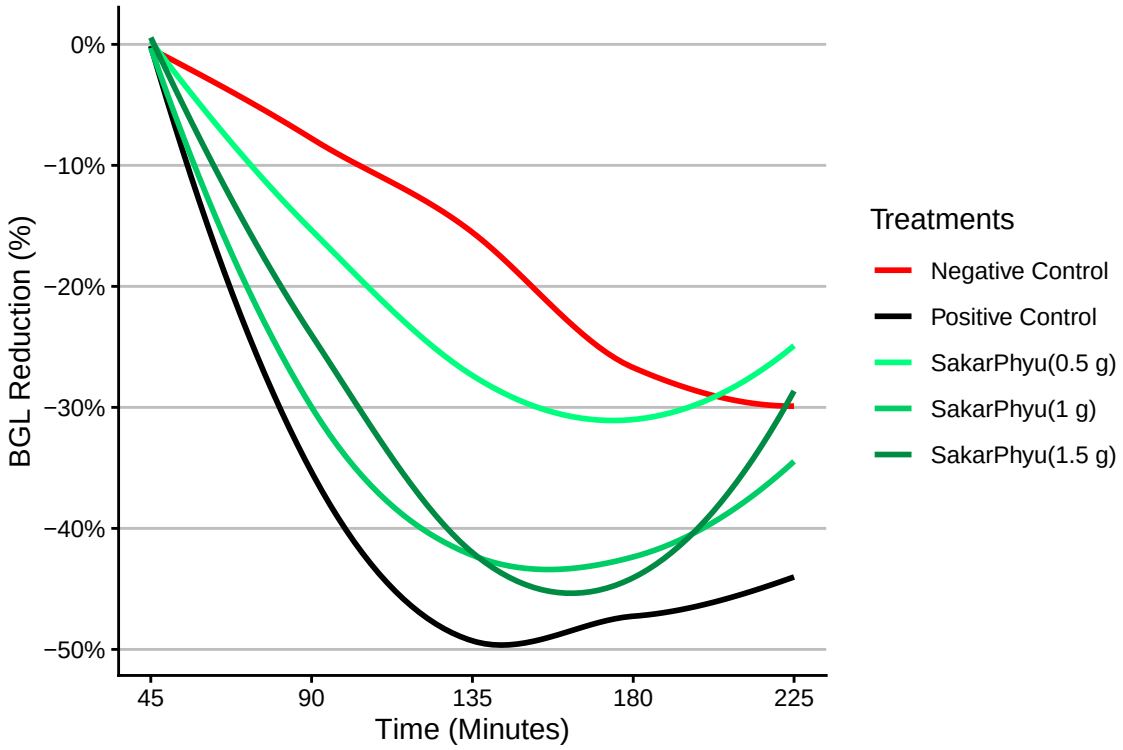
```r
df%>%
  ggplot(aes(time, Percent_reduction, color = Treatments))+
  geom_smooth(se=F, span =1)+
  #geom_point()+
  scale_color_manual(#breaks = trts,
                     values = c("red","black","blue",
                                "springgreen",
                                "springgreen3","springgreen4"))+
  scale_x_continuous(limits = c(45,225),
                     breaks = seq(45,225, 45))+
  scale_y_continuous(labels = scales::percent_format(scale = 1))+
  labs(x="Time (Minutes)",
       y="BGL Reduction (%)") +
  theme_classic()+
  theme(
    panel.grid.major.y = element_line(color = "grey", size = 0.5)
  )
```




```r
# or without "normal", which is data from no administration of both treatmens and BGL induction

df%>%filter(Treatments!="Normal") %>%
  ggplot(aes(time, Percent_reduction, color = Treatments))+
  geom_smooth(se=F, span =1)+
  scale_color_manual(#breaks = Treatments,
                     values = c("red","black",
                                "springgreen",
                                "springgreen3","springgreen4"))+
    scale_x_continuous(limits = c(45,225),
```

```
                    breaks = seq(45,225, 45))+
scale_y_continuous(labels = scales::percent_format(scale=1))+
labs(x="Time (Minutes)",
     y="BGL Reduction (%)") +
theme_classic()+
theme(
  panel.grid.major.y = element_line(color = "grey", size = 0.5)
)
```



Analysis of Variance (F-test)

In ANOVA, the comparison would be made on ***original BGL data*** (not on calculated reduction percent data) between the two controls and the treatments, excluding the "Normal", which is just a type of assessment of measurement system. The data at 0 min would be excluded from the analysis because they are measured before the administration and invalid for treatment comparisons.

```
df2<-Book1 %>% filter(Treatments !="Normal" & time !=0)
```

The following data frame, will be used for the further analysis

	Treatments	time	BGL
1	Negative Control	45	225
2	Negative Control	45	165
3	Negative Control	45	153
4	Negative Control	45	180
5	Negative Control	45	201

	Treatments	time	BGL
6	Negative Control	90	183
7	Negative Control	90	160
8	Negative Control	90	137
9	Negative Control	90	170
10	Negative Control	90	185
11	Negative Control	135	158
12	Negative Control	135	153
13	Negative Control	135	127
14	Negative Control	135	167
15	Negative Control	135	168
16	Negative Control	180	134
17	Negative Control	180	129
18	Negative Control	180	108
19	Negative Control	180	118
20	Negative Control	180	157
21	Negative Control	225	147
22	Negative Control	225	115
23	Negative Control	225	109
24	Negative Control	225	133
25	Negative Control	225	149
26	Positive Control	45	168
27	Positive Control	45	190
28	Positive Control	45	209
29	Positive Control	45	189
30	Positive Control	45	152
31	Positive Control	90	93
32	Positive Control	90	108
33	Positive Control	90	153
34	Positive Control	90	123
35	Positive Control	90	107
36	Positive Control	135	65
37	Positive Control	135	85
38	Positive Control	135	93
39	Positive Control	135	123
40	Positive Control	135	92
41	Positive Control	180	89
42	Positive Control	180	114
43	Positive Control	180	80
44	Positive Control	180	110
45	Positive Control	180	107
46	Positive Control	225	84
47	Positive Control	225	91
48	Positive Control	225	80
49	Positive Control	225	107
50	Positive Control	225	126
51	SakarPhyu(0.5 g)	45	180
52	SakarPhyu(0.5 g)	45	141
53	SakarPhyu(0.5 g)	45	189
54	SakarPhyu(0.5 g)	45	166
55	SakarPhyu(0.5 g)	45	190
56	SakarPhyu(0.5 g)	90	147
57	SakarPhyu(0.5 g)	90	130

	Treatments	time	BGL
58	SakarPhyu(0.5 g)	90	163
59	SakarPhyu(0.5 g)	90	110
60	SakarPhyu(0.5 g)	90	194
61	SakarPhyu(0.5 g)	135	104
62	SakarPhyu(0.5 g)	135	145
63	SakarPhyu(0.5 g)	135	142
64	SakarPhyu(0.5 g)	135	63
65	SakarPhyu(0.5 g)	135	170
66	SakarPhyu(0.5 g)	180	89
67	SakarPhyu(0.5 g)	180	140
68	SakarPhyu(0.5 g)	180	139
69	SakarPhyu(0.5 g)	180	72
70	SakarPhyu(0.5 g)	180	150
71	SakarPhyu(0.5 g)	225	105
72	SakarPhyu(0.5 g)	225	141
73	SakarPhyu(0.5 g)	225	150
74	SakarPhyu(0.5 g)	225	90
75	SakarPhyu(0.5 g)	225	159
76	SakarPhyu(1 g)	45	221
77	SakarPhyu(1 g)	45	157
78	SakarPhyu(1 g)	45	206
79	SakarPhyu(1 g)	45	238
80	SakarPhyu(1 g)	45	172
81	SakarPhyu(1 g)	90	118
82	SakarPhyu(1 g)	90	117
83	SakarPhyu(1 g)	90	105
84	SakarPhyu(1 g)	90	168
85	SakarPhyu(1 g)	90	164
86	SakarPhyu(1 g)	135	100
87	SakarPhyu(1 g)	135	133
88	SakarPhyu(1 g)	135	80
89	SakarPhyu(1 g)	135	117
90	SakarPhyu(1 g)	135	122
91	SakarPhyu(1 g)	180	121
92	SakarPhyu(1 g)	180	137
93	SakarPhyu(1 g)	180	80
94	SakarPhyu(1 g)	180	99
95	SakarPhyu(1 g)	180	119
96	SakarPhyu(1 g)	225	135
97	SakarPhyu(1 g)	225	141
98	SakarPhyu(1 g)	225	95
99	SakarPhyu(1 g)	225	93
100	SakarPhyu(1 g)	225	156
101	SakarPhyu(1.5 g)	45	122
102	SakarPhyu(1.5 g)	45	182
103	SakarPhyu(1.5 g)	45	149
104	SakarPhyu(1.5 g)	45	192
105	SakarPhyu(1.5 g)	45	166
106	SakarPhyu(1.5 g)	90	118
107	SakarPhyu(1.5 g)	90	113
108	SakarPhyu(1.5 g)	90	87
109	SakarPhyu(1.5 g)	90	172

	Treatments	time	BGL
110	SakarPhyu(1.5 g)	90	139
111	SakarPhyu(1.5 g)	135	74
112	SakarPhyu(1.5 g)	135	132
113	SakarPhyu(1.5 g)	135	67
114	SakarPhyu(1.5 g)	135	69
115	SakarPhyu(1.5 g)	135	126
116	SakarPhyu(1.5 g)	180	69
117	SakarPhyu(1.5 g)	180	113
118	SakarPhyu(1.5 g)	180	57
119	SakarPhyu(1.5 g)	180	91
120	SakarPhyu(1.5 g)	180	126
121	SakarPhyu(1.5 g)	225	63
122	SakarPhyu(1.5 g)	225	134
123	SakarPhyu(1.5 g)	225	64
124	SakarPhyu(1.5 g)	225	194
125	SakarPhyu(1.5 g)	225	145

```
anova1<-aov(BGL~time+Treatments, data = df2)
summary(anova1)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
time          1  55056   55056  55.771 1.49e-11 ***
Treatments    4  22458    5615   5.687 0.000318 ***
Residuals   119 117476     987
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova2<-aov(BGL~time*Treatments, data = df2)
summary(anova2)
```

```
                Df Sum Sq Mean Sq F value   Pr(>F)
time             1  55056   55056  54.745 2.43e-11 ***
Treatments       4  22458    5615   5.583 0.000383 ***
time:Treatments  4   1821     455   0.453 0.770250
Residuals      115 115655    1006
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova3<-aov(BGL~time+I(time^2)+Treatments, data = df2)
summary(anova3)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
time          1  55056   55056  71.555 8.42e-14 ***
I(time^2)     1  26683   26683  34.679 3.74e-08 ***
Treatments    4  22458    5615   7.297 2.76e-05 ***
Residuals   118  90792     769
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
AIC(anova1, anova2, anova3)
```

```
       df      AIC
anova1  7 1224.444
anova2 11 1230.491
anova3  8 1194.237
```

```r
BIC(anova1, anova2, anova3)
```

```
       df      BIC
anova1  7 1244.242
anova2 11 1261.602
anova3  8 1216.863
```

AIC (Akaike Information Criterion): Penalizes model complexity. Lower is better. BIC (Bayesian Information Criterion): More stringent than AIC. Lower is better. AIC, BIC values showed that **anova3** is the best of the three.

```r
library(car)
anova(anova1, anova3)
```

```
Analysis of Variance Table

Model 1: BGL ~ time + Treatments
Model 2: BGL ~ time + I(time^2) + Treatments
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    119 117476
2    118  90792  1     26683 34.679 3.744e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In this analysis, it is obvious that the function of time showed quadratic effect I(time^2) on BGL reduction. The effect of Treatments showed significance at the p value of less than 0.000. The last model "anova3" seem reasonable choice as proved by the model comparison of anova1 and anova3. Since there is no interaction effect between treatments and time as seen in model 'anova2', post hoc analysis will be focused on only differences among treatments.

Post hoc Analysis

Because the p-value for Treatmens showed the significant difference among the factor levels, the post hoc analyis using Tukey method will be conducted for pari-wise comparision. This will output the grouping of treatments of similiar effect, with different alphabets showing different effects. The grouping is performed at alpha = 0.05 significant level.

```r
library(agricolae)

Tuk<-HSD.test(anova3, trt = c("Treatments"), alpha = 0.05, group = TRUE)
print(Tuk)
```

```
$statistics
   MSerror  Df    Mean      CV      MSD
  769.4274 118 132.768 20.8925 21.73567

$parameters
   test     name.t ntr StudentizedRange alpha
  Tukey Treatments   5         3.917953  0.05

$means
                    BGL      std  r       se Min Max Q25 Q50 Q75
Negative Control 153.24 28.89244 25 5.547711 108 225 133 153 168
Positive Control 117.52 38.44706 25 5.547711  65 209  91 107 126
SakarPhyu(0.5 g) 138.76 36.11357 25 5.547711  63 194 110 142 163
SakarPhyu(1 g)   135.76 41.67961 25 5.547711  80 238 105 122 157
SakarPhyu(1.5 g) 118.56 42.83114 25 5.547711  57 194  74 122 145

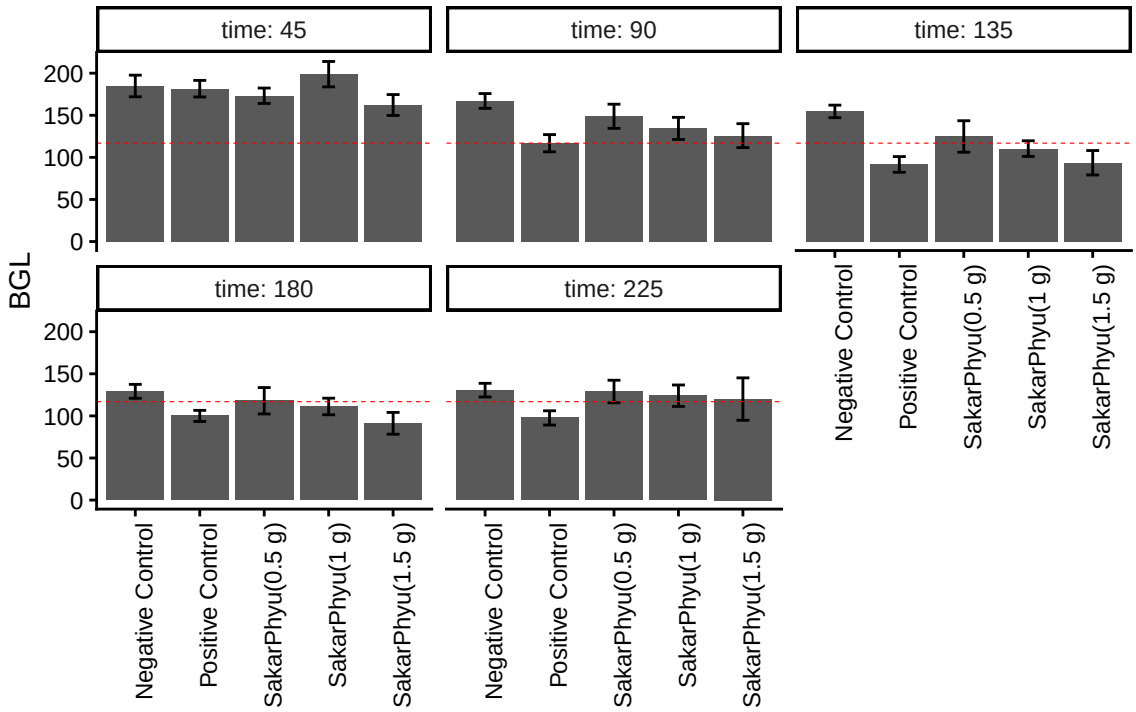
$comparison
NULL

$groups
                    BGL groups
Negative Control 153.24      a
SakarPhyu(0.5 g) 138.76     ab
SakarPhyu(1 g)   135.76     ab
SakarPhyu(1.5 g) 118.56      b
Positive Control 117.52      b

attr(,"class")
[1] "group"
```

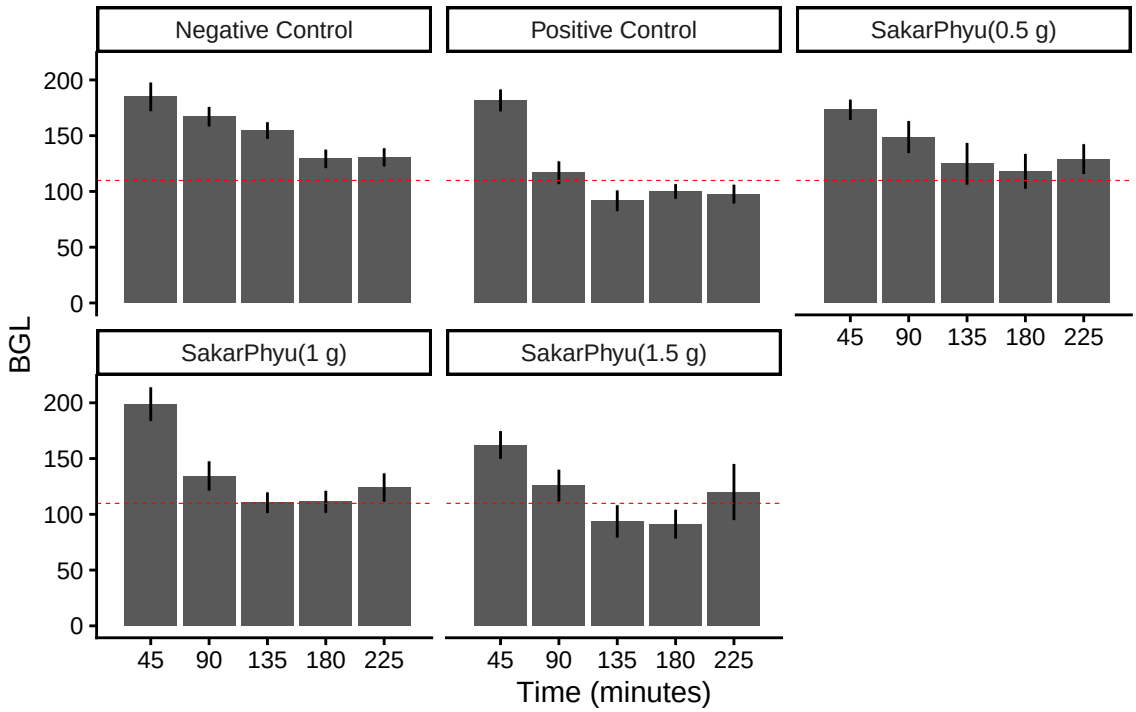
for more exploration

```
normal <- Book1 %>% filter(Treatments == "Normal" ) %>%
  summarise(meanbgl=mean(BGL)) %>% pull(meanbgl)
Book1 %>% filter(Treatments != "Normal" & time != 0) %>%
  ggplot(aes(Treatments, BGL))+
  geom_bar(stat='summary', fun = mean)+
  geom_errorbar(stat = "summary",fun.data = mean_se, width = 0.2)+
    geom_hline(yintercept = normal, color = "red", linewidth = 0.2,
               linetype = 'dashed')+
  facet_wrap(time~., labeller = "label_both")+
  labs(x=NULL,
       caption = paste0("Dashed red line indicate average of normal data (",round(normal,2),"), i.e., n
  theme_classic()+theme(axis.text.x = element_text(angle = 90, hjust=1))
```



Dashed red line indicate average of normal data (116.9), i.e., no treatment and no BGL induction

```
fastin <- fasting %>% summarise(mean=mean(BGL)) %>% pull(mean)
Book1 %>% filter(Treatments != "Normal" & time!=0) %>%
  ggplot(aes(time, BGL))+
  geom_bar(stat='summary', fun = mean)+
  geom_errorbar(stat = "summary",fun.data = mean_se, width = 0.1)+
  geom_hline(yintercept = fastin, color = "red", linewidth = 0.2,
             linetype = 'dashed')+
  labs(x="Time (minutes)",
       caption = paste0("Dashed red line indicate average of BGL (",round(fastin, 2),") before  treatmen
  scale_x_continuous(limits = c(15,250),
                     breaks = seq(45,250, 45))+
  facet_wrap(Treatments~.)+theme_classic()
```



Dashed red line indicate average of BGL (109.83) before treatment and BGL induction

Further analysis on Quadratic term of factor "time"

Since the model include the qudratic term, it is easier to interpret using the linear regression model. The estimate were in comparison with the default reference point represented as the intercept coefficient, which is the BGL value at the negative control and zero minute. The t-test of an estimate of interested treatments were performed to assess the significance of the difference from that reference value, adjusting for the effect of other factors, in this case, "time".

```
lm_mod <- lm(BGL~time+I(time^2)+Treatments, data = df2)
summary.lm(lm_mod)
```

```
Call:
lm(formula = BGL ~ time + I(time^2) + Treatments, data = df2)

Residuals:
    Min      1Q  Median      3Q     Max
-58.297 -21.669   0.617  23.097  87.657

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                  2.589e+02  1.289e+01  20.081  < 2e-16 ***
time                        -1.494e+00  2.015e-01  -7.414 2.03e-11 ***
I(time^2)                    4.312e-03  7.322e-04   5.889 3.74e-08 ***
TreatmentsPositive Control -3.572e+01  7.846e+00  -4.553 1.30e-05 ***
TreatmentsSakarPhyu(0.5 g) -1.448e+01  7.846e+00  -1.846   0.0675 .
TreatmentsSakarPhyu(1 g)   -1.748e+01  7.846e+00  -2.228   0.0278 *
```

```
TreatmentsSakarPhyu(1.5 g) -3.468e+01  7.846e+00  -4.420 2.20e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 27.74 on 118 degrees of freedom
Multiple R-squared:  0.5344,    Adjusted R-squared:  0.5107
F-statistic: 22.57 on 6 and 118 DF,  p-value: < 2.2e-16
```

Or, we can separately test the significance of the polynomial terms .

```
df2$time <- factor(df2$time)
contrasts(df2$time) <- contr.poly(5)
polymod <- aov(BGL~time+Treatments, data = df2)
summary(polymod)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
time          4  81761   20440  26.121 1.89e-15 ***
Treatments    4  22458    5615   7.175 3.38e-05 ***
Residuals   116  90771     783
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

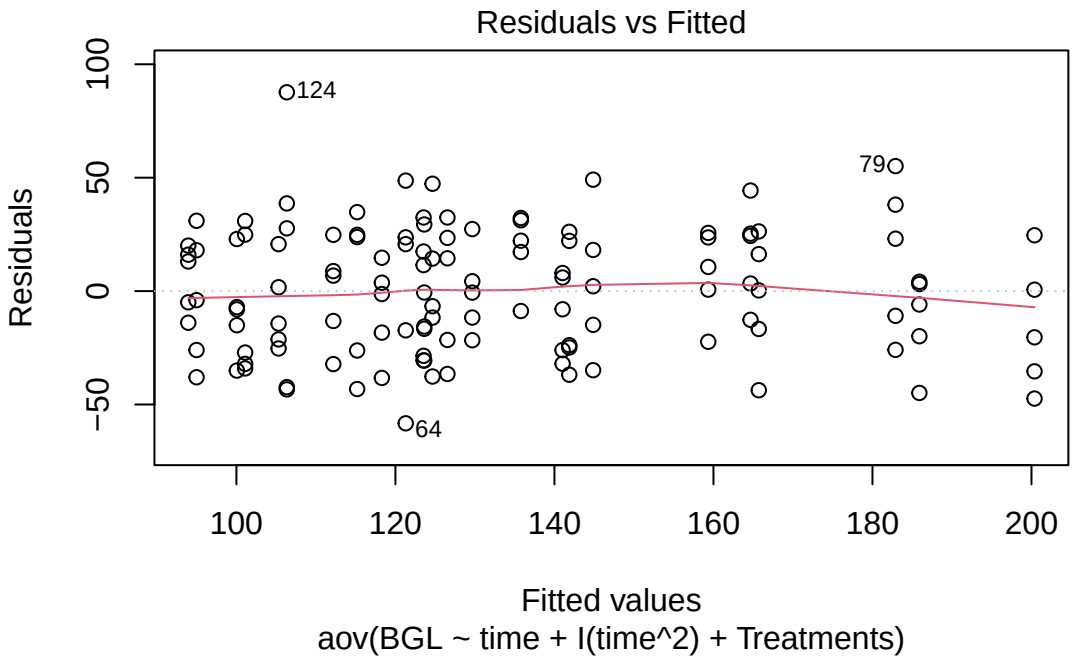
```
summary.aov(polymod, split =list(
  time = list("linear" = 1,
              "Quadratic" = 2)))
```

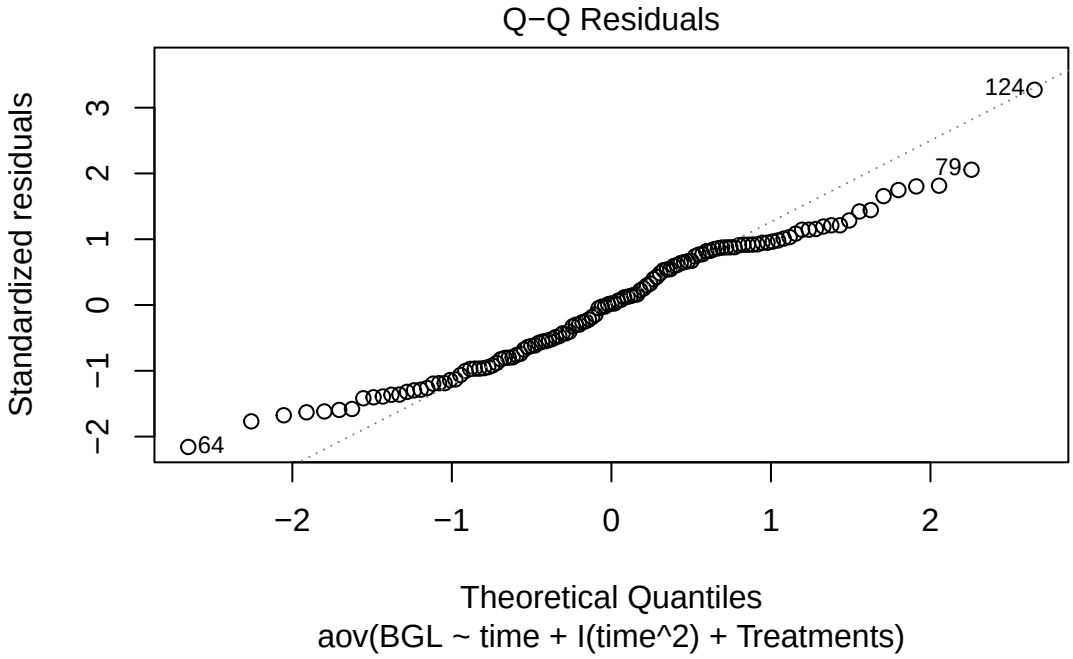
```
                  Df Sum Sq Mean Sq F value   Pr(>F)
time               4  81761   20440  26.121 1.89e-15 ***
  time: linear     1  55056   55056  70.359 1.36e-13 ***
  time: Quadratic  1  26683   26683  34.100 4.87e-08 ***
Treatments         4  22458    5615   7.175 3.38e-05 ***
Residuals        116  90771     783
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Here, we can clearly see the fit of quadratic term of time is significant, shown in both linear regression model and anova model with polynomial contrasts on factor "time".

Model Assumption Checking

```
plot(anova3, which = c(1,2))
```


Residuals vs Fitted
Residuals
100
50
0
−50
124
79
64
100
120
140
160
180
200
Fitted values
aov(BGL ~ time + I(time^2) + Treatments)


Q−Q Residuals
Standardized residuals
3
2
1
0
−1
−2
124
79
64
−2
−1
0
1
2
Theoretical Quantiles
aov(BGL ~ time + I(time^2) + Treatments)

Remodelling without potential outliers/influencers

Here we can see that three data points were very alarmingly not normal. may be outliers and influencers. these data points were:

```
df2[c(124, 79, 64),]
```

```
# A tibble: 3 x 3
  Treatments       time    BGL
  <chr>            <fct> <dbl>
1 SakarPhyu(1.5 g) 225     194
2 SakarPhyu(1 g)   45      238
3 SakarPhyu(0.5 g) 135      63
```

And next, conduct the analysis, again, with new data frame (without those 3 points.). Well, let's try removing the most prominent one first. i.e.,

rowname	Treatments	time	BGL
124	SakarPhyu(1.5 g)	225	194

, the 124th observation of the working data frame.

```
df3 <- df2[-c(124),]
polymod1 <- aov(BGL~time+Treatments, data = df3)
summary(polymod1)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
time          4  83649   20912  29.174  < 2e-16 ***
Treatments    4  25130    6282   8.765 3.27e-06 ***
Residuals   115  82432     717
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary.aov(polymod1, split =list(
  time = list("linear" = 1,
              "Quadratic" = 2)))
```

```
                   Df Sum Sq Mean Sq F value   Pr(>F)
time                4  83649   20912  29.174  < 2e-16 ***
  time: linear      1  59745   59745  83.349 2.77e-15 ***
  time: Quadratic   1  23808   23808  33.214 7.04e-08 ***
Treatments          4  25130    6282   8.765 3.27e-06 ***
Residuals         115  82432     717
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(polymod1)
```

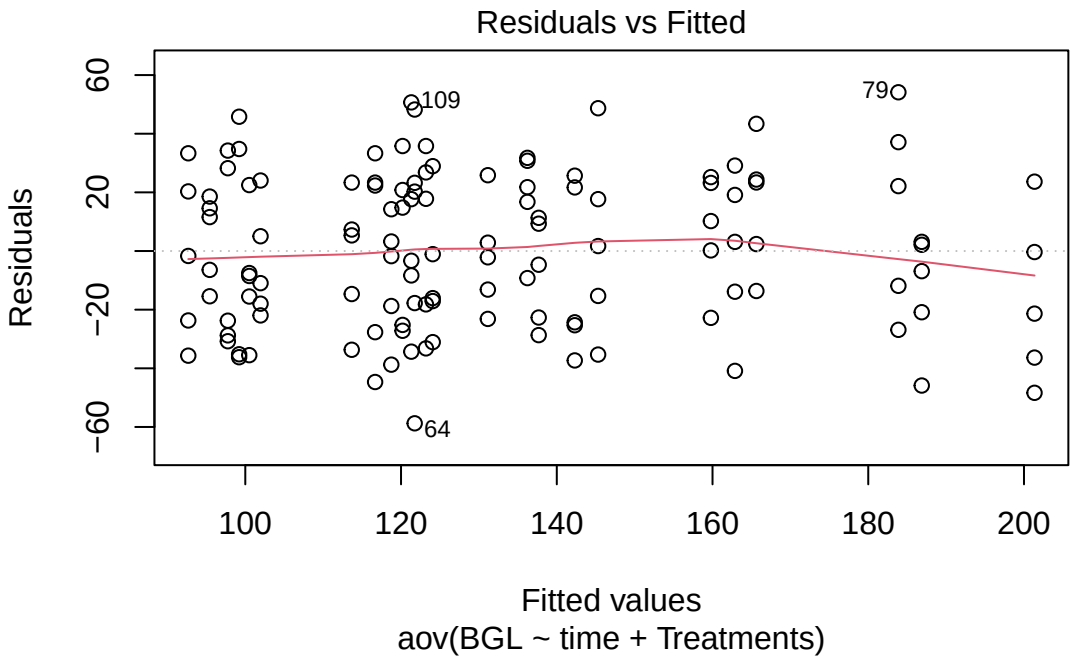
```
  Tukey multiple comparisons of means
    95% family-wise confidence level

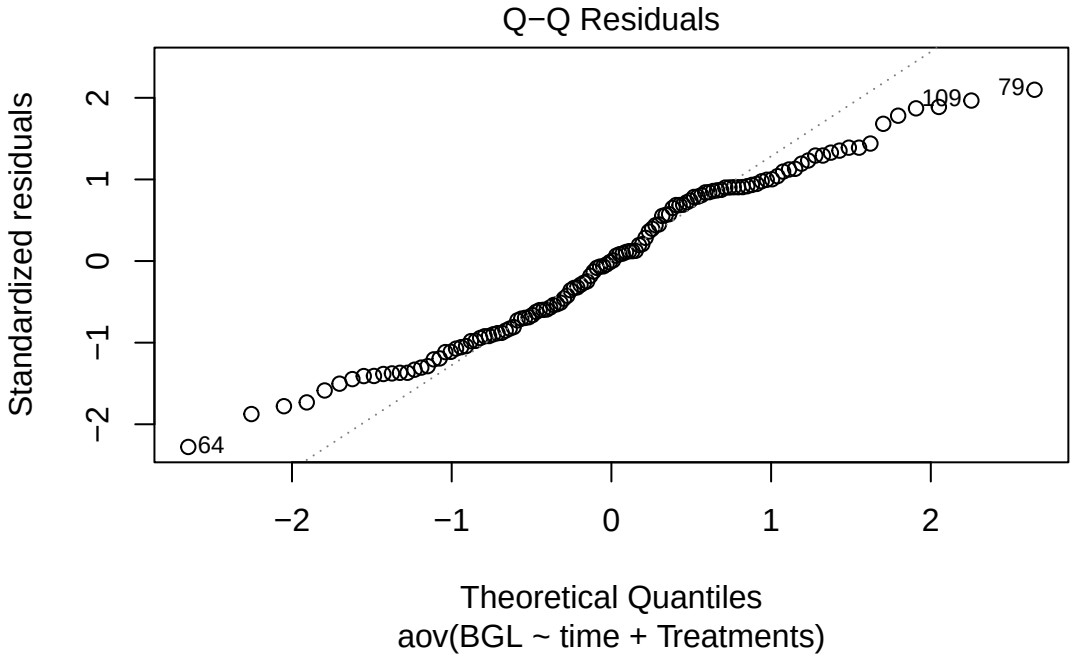
Fit: aov(formula = BGL ~ time + Treatments, data = df3)

$time
              diff       lwr         upr     p adj
90-45   -41.560000 -62.54771 -20.5722931 0.0000024
135-45  -65.120000 -86.10771 -44.1322931 0.0000000
180-45  -70.200000 -91.18771 -49.2122931 0.0000000
225-45  -62.953333 -84.15854 -41.7481314 0.0000000
135-90  -23.560000 -44.54771  -2.5722931 0.0195291
180-90  -28.640000 -49.62771  -7.6522931 0.0022720
225-90  -21.393333 -42.59854  -0.1881314 0.0468832
180-135  -5.080000 -26.06771  15.9077069 0.9622802
225-135   2.166667 -19.03854  23.3718686 0.9985738
225-180   7.246667 -13.95854  28.4518686 0.8777347

$Treatments
                                        diff         lwr         upr     p adj
Positive Control-Negative Control -35.720000 -56.7077069 -14.7322931 0.0000658
SakarPhyu(0.5 g)-Negative Control -14.480000 -35.4677069   6.5077069 0.3168050
SakarPhyu(1 g)-Negative Control   -17.480000 -38.4677069   3.5077069 0.1497121
SakarPhyu(1.5 g)-Negative Control -38.447778 -59.6529797 -17.2425759 0.0000182
SakarPhyu(0.5 g)-Positive Control  21.240000   0.2522931  42.2277069 0.0458177
SakarPhyu(1 g)-Positive Control    18.240000  -2.7477069  39.2277069 0.1203516
SakarPhyu(1.5 g)-Positive Control  -2.727778 -23.9329797  18.4774241 0.9964965
SakarPhyu(1 g)-SakarPhyu(0.5 g)    -3.000000 -23.9877069  17.9877069 0.9947340
SakarPhyu(1.5 g)-SakarPhyu(0.5 g) -23.967778 -45.1729797  -2.7625759 0.0183413
SakarPhyu(1.5 g)-SakarPhyu(1 g)   -20.967778 -42.1729797   0.2374241 0.0541848
```

```r
plot(polymod1, which = 1:2)
```





For another analysis, we'll exclude all three potential outliers, which are

rowname	Treatments	time	BGL
64	SakarPhyu(0.5 g)	135	63
79	SakarPhyu(1 g)	45	238
124	SakarPhyu(1.5 g)	225	194

.

```
df4 <- df2[-c(124, 79, 64),]
polymod2 <- aov(BGL~time+Treatments, data = df4)
summary(polymod2)
```

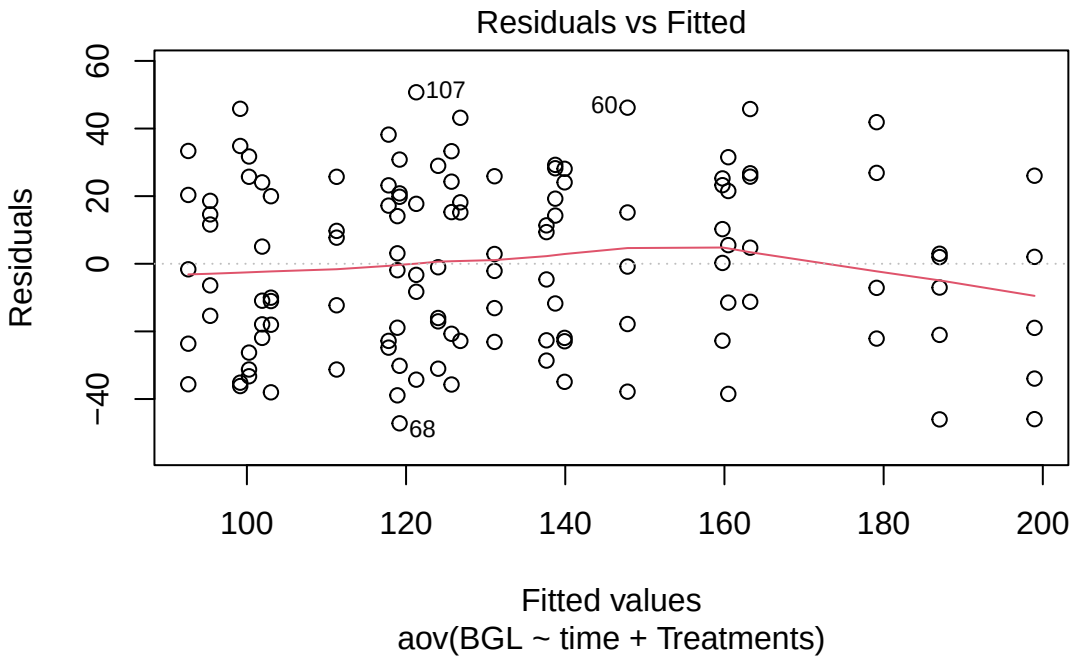
```
             Df Sum Sq Mean Sq F value   Pr(>F)
time          4  73967   18492  27.678 5.05e-16 ***
Treatments    4  25760    6440   9.639 9.62e-07 ***
Residuals   113  75496     668
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

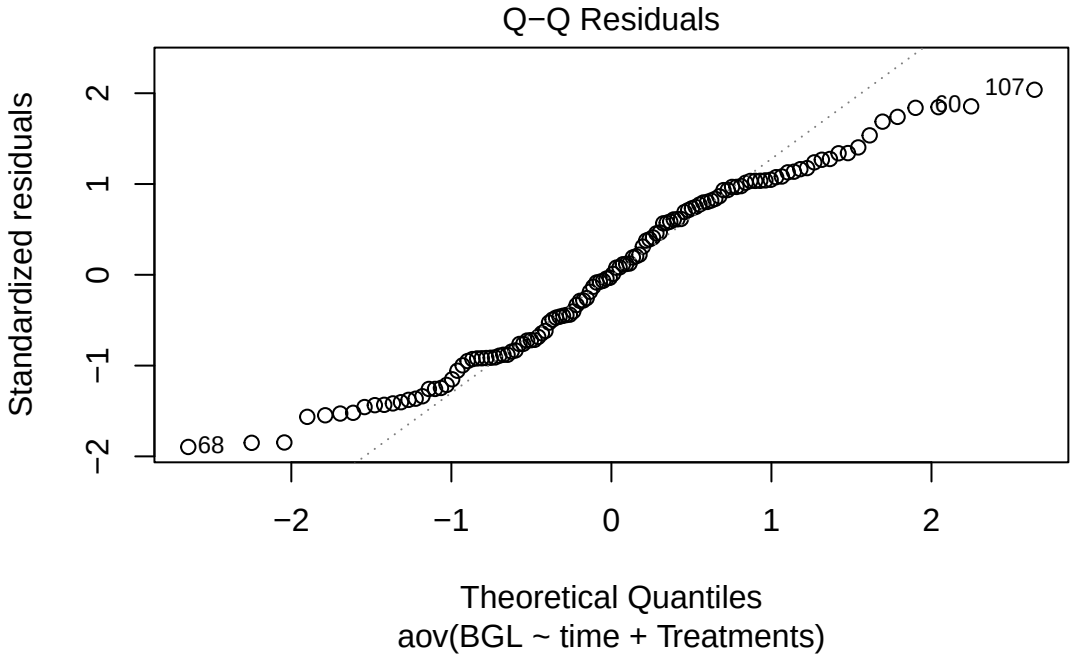
```
summary.aov(polymod2, split =list(
  time = list("linear" = 1,
              "Quadratic" = 2)))
```

```
                  Df Sum Sq Mean Sq F value   Pr(>F)
time               4  73967   18492  27.678 5.05e-16 ***
  time: linear     1  54210   54210  81.140 6.01e-15 ***
  time: Quadratic  1  19725   19725  29.524 3.21e-07 ***
Treatments         4  25760    6440   9.639 9.62e-07 ***
Residuals        113  75496     668
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

produce stronger statistical prove that there are differences among means of factor levels and p value for quadratic term also provide stronger prove.

```
plot(polymod2, which = 1:2)
```





residual vs fitted plot showed no violation of Homoscedasticity, which is more important for ANOVA than normality, which is seen in normal q-q plot and seemed violated.

```
Tuk<-HSD.test(polymod2, trt = c("Treatments"), alpha = 0.05, group = TRUE)
print(Tuk)
```

```
$statistics
   MSerror  Df     Mean       CV
  668.1053 113 131.9754 19.58526

$parameters
   test     name.t ntr StudentizedRange alpha
  Tukey Treatments   5         3.920649  0.05

$means
                      BGL      std  r       se Min Max    Q25   Q50    Q75
Negative Control 153.2400 28.89244 25 5.169547 108 225 133.00 153.0 168.00
Positive Control 117.5200 38.44706 25 5.169547  65 209  91.00 107.0 126.00
SakarPhyu(0.5 g) 141.9167 33.18056 24 5.276146  72 194 125.00 143.5 163.75
SakarPhyu(1 g)   131.5000 36.59651 24 5.276146  80 221 103.75 121.5 156.25
SakarPhyu(1.5 g) 115.4167 40.70030 24 5.276146  57 192  72.75 120.0 140.50

$comparison
NULL

$groups
                      BGL groups
Negative Control 153.2400      a
SakarPhyu(0.5 g) 141.9167     ab
SakarPhyu(1 g)   131.5000     bc
Positive Control 117.5200      c
SakarPhyu(1.5 g) 115.4167      c

attr(,"class")
[1] "group"
```

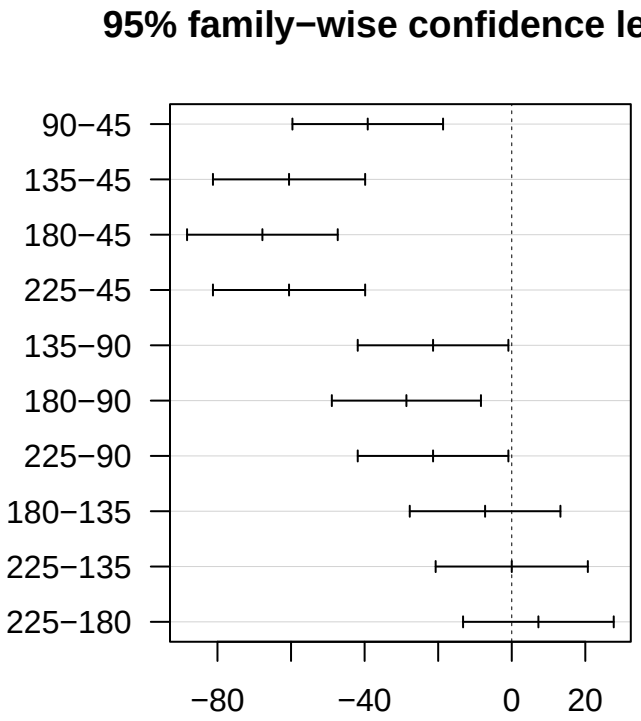
```
Tuk<-HSD.test(polymod2, trt = c("Treatments"), alpha = 0.05, group = F)
print(Tuk$comparison[,1:3])
```

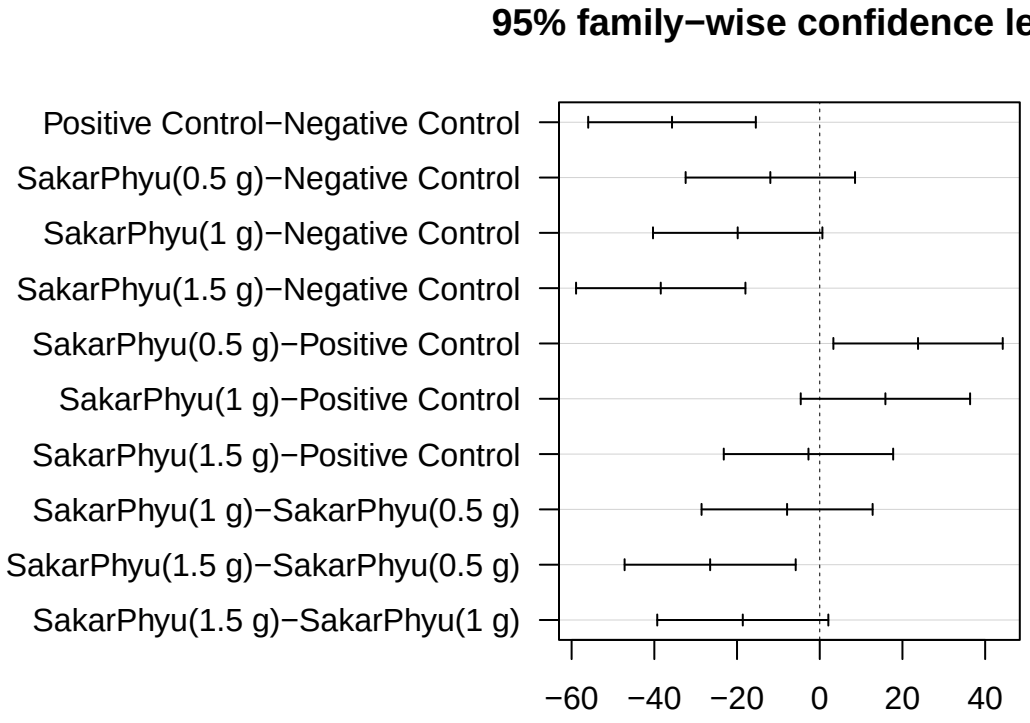
```
                                  difference pvalue signif.
Negative Control - Positive Control  35.720000 0.0000     ***
Negative Control - SakarPhyu(0.5 g)  11.323333 0.5434
Negative Control - SakarPhyu(1 g)    21.740000 0.0316       *
Negative Control - SakarPhyu(1.5 g)  37.823333 0.0000     ***
Positive Control - SakarPhyu(0.5 g) -24.396667 0.0110       *
Positive Control - SakarPhyu(1 g)   -13.980000 0.3273
Positive Control - SakarPhyu(1.5 g)   2.103333 0.9985
SakarPhyu(0.5 g) - SakarPhyu(1 g)    10.416667 0.6315
SakarPhyu(0.5 g) - SakarPhyu(1.5 g)  26.500000 0.0050      **
SakarPhyu(1 g) - SakarPhyu(1.5 g)    16.083333 0.2044
```

```
#TukeyHSD(polymod)
```

The results showed that positive control and sakarphyu (1.5 g) seem to have similar effect while negative control and sakarphyu (0.5 g) does the same. Sakarphyu (1 g) showed similiarity with positive control and sakarphyu (0.5 g)

```r
par(mar = c(3,15,3,3))
plot(TukeyHSD(polymod2), las = 1)
```





If the horizontal lines representing the mean difference and confident level of the two comparing factor levels did not cross the dashed zero vertical line, the difference between the two means is significant at alpha = 0.05. If the line cross it, the difference is not significant.

which model to use?

```
Tuk <- HSD.test(polymod1, "Treatments", group = F)
p1 <- Tuk$comparison[,2:3]
Tuk <- HSD.test(polymod2, "Treatments", group = F)
p2 <- Tuk$comparison[,2:3]
Tuk <- HSD.test(anova3, "Treatments", group = F)
p3 <- Tuk$comparison[,2:3]

P123 <- cbind(p1,p2,p3)
names(P123) <- c("polymod1","sig1","polymod2","sig2","anova3","sig3")
#p1 <- TukeyHSD(polymod1)$Treatments[, 4]
#p2 <- TukeyHSD(polymod2)$Treatments[, 4]
#p3 <- TukeyHSD(anova3)$Treatments[, 4]

# comparisondf <- data.frame(polymod_1=p1,polymod_2=p2, anova_3=p3)
# # p values calculated by Tukey test for selected 3 models
# round(comparisondf,4)
#
# comparisondf %>% mutate("Pair-wise Comparison"=rownames(.)) %>%
# pivot_longer(cols = 1:3, names_to = "mods", values_to = "p")%>%
#   mutate(s=ifelse(p<0.001, "***", ifelse(p<0.01, "**",ifelse(p<0.05, "*", "NS")))) %>%
#   select(!"p") %>%
#   pivot_wider(names_from = mods, values_from = s) %>% print(n=Inf)

AIC(polymod1,polymod2,anova3)
```

```
         df      AIC
polymod1 10 1177.829
polymod2 10 1150.414
anova3    8 1194.237
```

```
BIC(polymod1,polymod2,anova3)
```

```
         df      BIC
polymod1 10 1206.031
polymod2 10 1178.454
anova3    8 1216.863
```

	polymod1	sig1	polymod2	sig2	anova3	sig3
Negative Control - Positive Control	0.0001	***	0.0000	***	0.0001	***
Negative Control - SakarPhyu(0.5 g)	0.3168		0.5434		0.3527	
Negative Control - SakarPhyu(1 g)	0.1497		0.0316	*	0.1767	
Negative Control - SakarPhyu(1.5 g)	0.0000	***	0.0000	***	0.0002	***

	polymod1	sig1	polymod2	sig2	anova3	sig3
Positive Control - SakarPhyu(0.5 g)	0.0458	*	0.0110	*	0.0588	.
Positive Control - SakarPhyu(1 g)	0.1204		0.3273		0.1443	
Positive Control - SakarPhyu(1.5 g)	0.9987		0.9985		0.9999	
SakarPhyu(0.5 g) - SakarPhyu(1 g)	0.9947		0.6315		0.9954	
SakarPhyu(0.5 g) - SakarPhyu(1.5 g)	0.0233	*	0.0050	**	0.0817	.
SakarPhyu(1 g) - SakarPhyu(1.5 g)	0.0666	.	0.2044		0.1898	

According to these analysis, model "polymod2" is the best of the three based on having lowest values in both AIC and BIC. Revisiting what the polymod2 is, it is build without 3 potential ourliers. in post hoc analysis with Tukey test, it showed the best resolution on comparing the Treatments effect.

time as random factor and treatment as fixed factor

```
library(lme4)
library(lmerTest)
lmmod<-lmer(BGL~Treatments+(1|time), data = df2)
summary(lmmod)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: BGL ~ Treatments + (1 | time)
   Data: df2

REML criterion at convergence: 1169.2

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.09743 -0.77968  0.05464  0.81040  3.12756

Random effects:
 Groups   Name        Variance Std.Dev.
 time     (Intercept) 786.3    28.04
 Residual             782.5    27.97
Number of obs: 125, groups:  time, 5

Fixed effects:
                           Estimate Std. Error      df t value Pr(>|t|)
(Intercept)                 153.240     13.732   5.315  11.159 6.87e-05 ***
TreatmentsPositive Control  -35.720      7.912 116.000  -4.515 1.53e-05 ***
TreatmentsSakarPhyu(0.5 g)  -14.480      7.912 116.000  -1.830   0.0698 .
TreatmentsSakarPhyu(1 g)    -17.480      7.912 116.000  -2.209   0.0291 *
TreatmentsSakarPhyu(1.5 g)  -34.680      7.912 116.000  -4.383 2.58e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmPC TSP(0g TSP(1g)
TrtmntsPstC -0.288
TrtSP(0.5g) -0.288  0.500
```

```
TrtmnSP(1g) -0.288  0.500  0.500
TrtSP(1.5g) -0.288  0.500  0.500  0.500
```

```
library(emmeans)
posthoc_results <- emmeans(lmmod, pairwise ~ Treatments, adjust = "tukey")
print(posthoc_results)
```

```
$emmeans
 Treatments       emmean   SE   df lower.CL upper.CL
 Negative Control    153 13.7 5.31    118.6      188
 Positive Control    118 13.7 5.31     82.8      152
 SakarPhyu(0.5 g)    139 13.7 5.31    104.1      173
 SakarPhyu(1 g)      136 13.7 5.31    101.1      170
 SakarPhyu(1.5 g)    119 13.7 5.31     83.9      153

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
 contrast                              estimate   SE  df t.ratio p.value
 Negative Control - Positive Control    35.72 7.91 116   4.515  0.0001
 Negative Control - SakarPhyu(0.5 g)    14.48 7.91 116   1.830  0.3614
 Negative Control - SakarPhyu(1 g)      17.48 7.91 116   2.209  0.1836
 Negative Control - SakarPhyu(1.5 g)    34.68 7.91 116   4.383  0.0002
 Positive Control - SakarPhyu(0.5 g)   -21.24 7.91 116  -2.685  0.0624
 Positive Control - SakarPhyu(1 g)     -18.24 7.91 116  -2.305  0.1506
 Positive Control - SakarPhyu(1.5 g)    -1.04 7.91 116  -0.131  0.9999
 SakarPhyu(0.5 g) - SakarPhyu(1 g)       3.00 7.91 116   0.379  0.9956
 SakarPhyu(0.5 g) - SakarPhyu(1.5 g)    20.20 7.91 116   2.553  0.0862
 SakarPhyu(1 g) - SakarPhyu(1.5 g)      17.20 7.91 116   2.174  0.1970

Degrees-of-freedom method: kenward-roger
P value adjustment: tukey method for comparing a family of 5 estimates
```

Lastly Session Info

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13)
Platform: x86_64-pc-linux-gnu
Running under: EndeavourOS

Matrix products: default
BLAS:   /usr/lib/libblas.so.3.12.0
LAPACK: /usr/lib/liblapack.so.3.12.0  LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
```

```
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Asia/Yangon
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] emmeans_1.11.2-8 lmerTest_3.1-3   lme4_1.1-37      Matrix_1.7-3
 [5] agricolae_1.3-7  car_3.1-3        carData_3.0-5    lubridate_1.9.4
 [9] forcats_1.0.0    stringr_1.5.2    dplyr_1.1.4      purrr_1.1.0
[13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_4.0.0
[17] tidyverse_2.0.0  readxl_1.4.5

loaded via a namespace (and not attached):
 [1] gtable_0.3.6        xfun_0.53           lattice_0.22-7
 [4] tzdb_0.5.0          numDeriv_2016.8-1.1 vctrs_0.6.5
 [7] tools_4.5.1         Rdpack_2.6.4        generics_0.1.4
[10] parallel_4.5.1      pbkrtest_0.5.5      cluster_2.1.8.1
[13] AlgDesign_1.2.1.2   pkgconfig_2.0.3     RColorBrewer_1.1-3
[16] S7_0.2.0            lifecycle_1.0.4     compiler_4.5.1
[19] farver_2.1.2        htmltools_0.5.8.1   yaml_2.3.10
[22] Formula_1.2-5       pillar_1.11.1       nloptr_2.2.1
[25] MASS_7.3-65         reformulas_0.4.1    boot_1.3-31
[28] abind_1.4-8         nlme_3.1-168        tidyselect_1.2.1
[31] digest_0.6.37       mvtnorm_1.3-3       stringi_1.8.7
[34] labeling_0.4.3      splines_4.5.1       fastmap_1.2.0
[37] grid_4.5.1          cli_3.6.5           magrittr_2.0.4
[40] utf8_1.2.6          broom_1.0.10        withr_3.0.2
[43] backports_1.5.0     scales_1.4.0        estimability_1.5.1
[46] timechange_0.3.0    rmarkdown_2.29      cellranger_1.1.0
[49] hms_1.1.3           evaluate_1.0.5      knitr_1.50
[52] rbibutils_2.3       mgcv_1.9-3          rlang_1.1.6
[55] Rcpp_1.1.0          xtable_1.8-4        glue_1.8.0
[58] rstudioapi_0.17.1   minqa_1.2.8         R6_2.6.1
```

References

[1] M.M. Syeed, M.S. Hossain, M.R. Karim, M.F. Uddin, M. Hasan, R.H. Khan, Surface water quality profiling using the water quality index, pollution index and statistical methods: A critical review, Environmental and Sustainability Indicators 18 (2023) 100247.

[2] S. Dash, A.S. Kalamdhad, Science mapping approach to critical reviewing of published literature on water quality indexing, Ecological Indicators 128 (2021) 107862.

[3] H. Wickham, M. Averick, J. Bryan, W. Chang, L.D. McGowan, R. François, G. Grolemund, A. Hayes, L. Henry, J. Hester, M. Kuhn, T.L. Pedersen, E. Miller, S.M. Bache, K. Müller, J. Ooms, D. Robinson, D.P. Seidel, V. Spinu, K. Takahashi, D. Vaughan, C. Wilke, K. Woo, H. Yutani, Welcome to the tidyverse, Journal of Open Source Software 4 (2019) 1686. https://doi.org/10.21105/joss.01686.

[4] H. Wickham, ggplot2: Elegant graphics for data analysis, Springer-Verlag New York, 2016. https://ggplot2.tidyverse.org.